TP LSQL DA2I L2 EMIT EMIT FIANARANTSOA (2025-2026)

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A- QCM

1- SQL est l'acronyme de **Structured Query Language**

2- Le mot-clé SQL utilisé pour rechercher des données dans une base de donnée est **SELECT**

3- Le mot-clé SQL utilisé pour modifier des données dans une base de donnée est **UPDATE**

4- Le mot-clé SQL utilisé pour ajouter de nouvelles données dans une base de donnée est **INSERT**

5- Le mot-clé SQL utilisé pour supprimer des données dans une base de donnée est **DELETE**

6- Le(s) mot(s)-clé(s) SQL utilisé(s) pour trier les résultats d'une requête est ORDER BY, **ORDER**

B- TRAVAUX PRATIQUES

TP1 LES JEUX OLYMPIQUES

1- A)

```
sqlite> SELECT *
   ...> FROM Medals
   ...> WHERE Year = 2008 AND Country = 'France';
```

Id	City	Year	Sport	Discipline	Event	Athlete	Gender	Country_Code	Country	Event_gender	Medal
13453	Beijing	2008	Aquatics	Swimming	4x100m freestyle relay	GILOT, Fabien	Men	FRA	France	M	Silver
13458	Beijing	2008	Aquatics	Swimming	4x100m freestyle relay	LEVEAUX, Amaury	Men	FRA	France	M	Silver
13478	Beijing	2008	Aquatics	Swimming	50m freestyle	BERNARD, Alain	Men	FRA	France	M	Bronze
13507	Beijing	2008	Aquatics	Swimming	4x100m freestyle relay	STEIMETZ, Boris	Men	FRA	France	M	Silver
13515	Beijing	2008	Aquatics	Swimming	4x100m freestyle relay	BOUSQUET, Frederick	Men	FRA	France	M	Silver
13573	Beijing	2008	Aquatics	Swimming	100m freestyle	BERNARD, Alain	Men	FRA	France	M	Gold
13577	Beijing	2008	Aquatics	Swimming	4x100m freestyle relay	MALLET, Gregory	Men	FRA	France	M	Silver
13579	Beijing	2008	Aquatics	Swimming	50m freestyle	LEVEAUX, Amaury	Men	FRA	France	M	Silver
13582	Beijing	2008	Aquatics	Swimming	4x100m freestyle relay	BERNARD, Alain	Men	FRA	France	M	Silver
13615	Beijing	2008	Aquatics	Swimming	200m breaststroke	DUBOSCQ, Hugues	Men	FRA	France	M	Bronze
13623	Beijing	2008	Aquatics	Swimming	100m breaststroke	DUBOSCQ, Hugues	Men	FRA	France	M	Bronze
13749	Beijing	2008	Archery	Archery	team (FITA Olympic round - 70m)	ARNOLD, Virginie	Women	FRA	France	W	Bronze
13756	Beijing	2008	Archery	Archery	team (FITA Olympic round - 70m)	SCHUH, Berengere	Women	FRA	France	W	Bronze
13761	Beijing	2008	Archery	Archery	team (FITA Olympic round - 70m)	DODEMONT, Sophie	Women	FRA	France	W	Bronze
13852	Beijing	2008	Athletics	Athletics	1500m	BAALA, Mehdi	Men	FRA	France	M	Bronze
13921	Beijing	2008	Athletics	Athletics	3000m steeplechase	MEKHISSI-B., Mahiedine	Men	FRA	France	M	Silver
14111	Beijing	2008	Boxing	Boxing	57 - 60kg (lightweight)	SOW, Daouda	Men	FRA	France	M	Silver
14115	Beijing	2008	Boxing	Boxing	60 - 64 kg	VASTINE, Alexis	Men	FRA	France	M	Bronze
14124	Beijing	2008	Boxing	Boxing	54 - 57kg (featherweight)	DJELKHIR, Khedafi	Men	FRA	France	M	Silver
14159	Beijing	2008	Canoe / Kayak	Canoe / Kayak F	K-2 500m (kayak double)	DELATTRE, Marie	Women	FRA	France	W	Bronze
14188	Beijing	2008	Canoe / Kayak	Canoe / Kayak F	K-2 500m (kayak double)	VIARD, Anne-Laure	Women	FRA	France	W	Bronze
14223	Beijing	2008	Canoe / Kayak	Canoe / Kayak S	K-1 (kayak single)	LEFEVRE, Fabien	Men	FRA	France	M	Silver
14237	Beijing	2008	Cycling	BMX	individual	LE CORGUILLE, Laetitia	Women	FRA	France	W	Silver
14238	Beijing	2008	Cycling	BMX	individual	CHAUSSON, Anne-Caroline	Women	FRA	France	W	Gold
14263	Beijing	2008	Cycling	Cycling Track	Team Sprint	SIREAU, Kevin	Men	FRA	France	M	Silver
14268	Beijing	2008	Cycling	Cycling Track	Sprint indivual	BOURGAIN, Mickael	Men	FRA	France	M	Bronze
14281	Beijing	2008	Cycling	Cycling Track	Team Sprint	TOURNANT, Arnaud	Men	FRA	France	M	Silver
14290	Beijing	2008	Cycling	Cycling Track	Team Sprint	BAUGE, Gregory	Men	FRA	France	M	Silver
14301	Beijing	2008	Cycling	Mountain Bike	cross-country	ABSALON, Julien	Men	FRA	France	M	Gold
14302	Beijing	2008	Cycling	Mountain Bike	cross-country	PERAUD, Jean-Christophe	Men	FRA	France	M	Silver
14355	Beijing	2008	Fencing	Fencing	épée team	JEANNET, Jerome	Men	FRA	France	M	Gold
14360	Beijing	2008	Fencing	Fencing	sabre team	LOPEZ, Nicolas	Men	FRA	France	M	Gold
14375	Beijing	2008	Fencing	Fencing	sabre individual	LOPEZ, Nicolas	Men	FRA	France	M	Silver
14376	Beijing	2008	Fencing	Fencing	sabre team	SANSON, Boris	Men	FRA	France	M	Gold
14393	Beijing	2008	Fencing	Fencing	épée team	ROBEIRI, Ulrich	Men	FRA	France	M	Gold
14402	Beijing	2008	Fencing	Fencing	épée individual	JEANNET, Fabrice	Men	FRA	France	M	Silver

Ce code affiche toutes les informations de la table **Medals** concernant les médailles remportées par la **France** durant l'année **2008**.

B)

```
sqlite> SELECT City, Year, Athlete FROM Medals WHERE Medal='Gold' AND Event='110m hurdles';
```

City	Year	Athlete
Montreal	1976	DRUT, Guy
Moscow	1980	MUNKELT, Thomas
Los Angeles	1984	KINGDOM, Roger
Seoul	1988	KINGDOM, Roger
Barcelona	1992	MCKOY, Mark
Atlanta	1996	JOHNSON, Allen
Sydney	2000	GARCIA, Anier
Athens	2004	LIU, Xiang
Beijing	2008	ROBLES, Dayron

Ce code affiche les villes, les années et les athlètes qui ont gagné une médaille d'or dans l'épreuve du 110 mètres haies dans la table **Medals**.

C)

```
sqlite> SELECT Athlete, Event FROM Medals WHERE Medal='Gold' AND Country='France' AND Year=2000 ORDER BY Athlete ASC;
```

Athlete	Event
ANDRIEUX, Michel	coxless pair (2-)
ASLOUM, Brahim	- 48kg (light-flyweight)
BALLANGER, Felicia	500m time trial
BALLANGER, Felicia	sprint
BETTE, Jean-Christophe	lightweight coxless four (4-)
DORFMANN, Xavier	lightweight coxless four (4-)
DOUILLET, David	+ 100kg (heavyweight)
DUMOULIN, Franck	10m air pistol (60 shots)
ESTANGUET, Tony	C-1 (canoe single)
FERRARI, Jean-Noel	foil team
GANE, Laurent	Olympic Sprint
GUYART, Brice	foil team
HOCDE, Yves	lightweight coxless four (4-)
LHOTELLIER, Patrice	foil team
MARTINEZ, Miguel	cross-country
PLUMENAIL, Lionel	foil team
PORCHIER, Laurent	lightweight coxless four (4-)
ROLLAND, Jean Christophe	coxless pair (2-)
ROUSSEAU, Florian	Olympic Sprint
ROUSSEAU, Florian	Keirin
TOURNANT, Arnaud	Olympic Sprint
VANDENHENDE, Severine	57 - 63kg (half-middleweight)

Cette requête affiche les athlètes et leurs épreuves ayant remporté une médaille d'or pour la France en 2000, en classant les résultats par ordre alphabétique des noms d'athlètes.

D)

```
sqlite> SELECT DISTINCT Country FROM Medals WHERE Medal='Gold' AND YEAR='1984' ORDER BY Country DESC;
```

Country
Yugoslavia
West Germany
United States
United Kingdom
Sweden
Spain
Romania
Portugal
Pakistan
New Zealand
Netherlands
Morocco
Mexico
Korea, South
Kenya
Japan
Italy
France
Finland
China
Canada
Brazil
Belgium
Austria
Australia

Cette requête affiche la liste des pays ayant remporté au moins une médaille d’or en 1984, sans doublons, triés par ordre alphabétique inverse.

E)

```
sqlite> SELECT Athlete, City, Year, Medal, Country FROM Medals WHERE Country LIKE '%nia';
```

Athlete	City	Year	Medal	Country
MEGELEA, Gheorghe	Montreal	1976	Bronze	Romania
DAFINOIU, Costica	Montreal	1976	Bronze	Romania
SIMON, Mircea	Montreal	1976	Silver	Romania
ZILBERMAN, Victor	Montreal	1976	Bronze	Romania
CUTOV, Simion	Montreal	1976	Silver	Romania
NASTAC, Alexandru	Montreal	1976	Bronze	Romania
DIBA, Vasile	Montreal	1976	Bronze	Romania
DIBA, Vasile	Montreal	1976	Gold	Romania
SERHEI, Larion	Montreal	1976	Bronze	Romania
SIMIONOV, Gheorghe	Montreal	1976	Silver	Romania
MALIHIN, Policarp	Montreal	1976	Bronze	Romania
DANIELOV, Gheorghe	Montreal	1976	Silver	Romania
NILCA, Alexandru	Montreal	1976	Bronze	Romania
POP, Ioan	Montreal	1976	Bronze	Romania
MARIN, Corneliu	Montreal	1976	Bronze	Romania
IRIMICIUC, Daniel	Montreal	1976	Bronze	Romania
MUSTATA, Marin	Montreal	1976	Bronze	Romania
CONSTANTIN, Mariana	Montreal	1976	Silver	Romania
UNGUREANU, Teodora	Montreal	1976	Silver	Romania
TRUSCA, Gabriela	Montreal	1976	Silver	Romania
GABOR, Georgeta	Montreal	1976	Silver	Romania
COMANECI, Nadia	Montreal	1976	Gold	Romania
GRECU, Danut	Montreal	1976	Bronze	Romania
GRIGORAS, Anca	Montreal	1976	Silver	Romania
UNGUREANU, Teodora	Montreal	1976	Bronze	Romania
COMANECI, Nadia	Montreal	1976	Bronze	Romania
UNGUREANU, Teodora	Montreal	1976	Silver	Romania
COMANECI, Nadia	Montreal	1976	Gold	Romania
COMANECI, Nadia	Montreal	1976	Silver	Romania
COMANECI, Nadia	Montreal	1976	Gold	Romania
COSMA, Adrian	Montreal	1976	Silver	Romania
PENU, Cornel	Montreal	1976	Silver	Romania
VOINEA, Radu	Montreal	1976	Silver	Romania
FOLKER, Alexandru	Montreal	1976	Silver	Romania
STOCKL, Werner	Montreal	1976	Silver	Romania
GRABOVSCI, Mircea	Montreal	1976	Silver	Romania
TUDOSIE, Constantin	Montreal	1976	Silver	Romania
GATU, Cristian	Montreal	1976	Silver	Romania
GUNES, Roland	Montreal	1976	Silver	Romania
LICU, Ghita	Montreal	1976	Silver	Romania
MUNTEANU, Nicolae	Montreal	1976	Silver	Romania
BIRTALAN, Stefan	Montreal	1976	Silver	Romania

Cette requête affiche les athlètes, villes, années, médailles et pays pour lesquels le nom du pays se termine par “nia”.

2) a)

```
sqlite> SELECT DISTINCT Athlete
   ...> FROM Medals
   ...> WHERE Country = 'France'
   ...>   AND Medal = 'Gold'
   ...>   AND Year = 1984;
```

Athlete
QUINON, Pierre
LAMOUR, Jean-Francois
BOISSE, Philippe
BIBARD, Michel
RUST, Albert
GARANDE, Patrice
ZANON, Jean Louis
BRISSON, Francois
CUBAYNES, Patrick
BENSOUSSAN, Michel
THOUVENEL, Jean Christophe
LACOMBE, Guy
LEMOULT, Jean-Claude
AYACHE, William
BIJOTAT, Dominique
SENAC, Didier
JEANNOL, Philippe
TOURE, Jose
XUEREB, Daniel
ROHR, Jean Philippe
HEBERLE, Philippe

b)

```
sqlite> SELECT Athlete, Year
   ...> FROM Medals
   ...> WHERE Event = 'marathon'
   ...>   AND Gender = 'Women'
   ...>   AND Medal = 'Gold'
   ...> ORDER BY Year ASC;
```

Athlete	Year
BENOIT, Joan	1984
MOTA, Rosa	1988
YEGOROVA, Valentina	1992
ROBA, Fatuma	1996
TAKAHASHI, Naoko	2000
NOGUCHI, Mizuki	2004
TOMESCU, Constantina	2008

c)

```
sqlite> SELECT Athlete, Medal, Discipline
   ...> FROM Medals WHERE Athlete LIKE '%, Majolie'
   ...> OR Athlete LIKE '%, Majolie';
```

d)

```
sqlite> SELECT Event
   ...> FROM Medals
   ...> WHERE Athlete = 'PHELPS, Michael'
   ...>   AND Year = 2004;
```

Event
100m butterfly
200m individual medley
400m individual medley
4x200m freestyle relay
4x100m freestyle relay
4x100m medley relay
200m butterfly
200m freestyle

e)

```
sqlite> SELECT Medal, COUNT(*) AS Total
   ...> FROM Medals
   ...> WHERE Country = 'Australia'
   ...>   AND Discipline = 'Swimming'
   ...>   AND (Medal = 'Gold' OR Medal = 'Silver')
   ...> GROUP BY Medal;
```

Medal	Total
Gold	61
Silver	72

3)

```
sqlite> SELECT Country, COUNT(*) FROM Medals WHERE Discipline='Swimming' AND Medal='Gold' GROUP BY Country ORDER BY COUNT(*) DESC;
```

Country	COUNT(*)
United States	255
Australia	61
East Germany	49
Netherlands	16
Soviet Union	14
Hungary	14
Unified team	11
Japan	7
China	7
South Africa	6
United Kingdom	5
Russia	5
Canada	5
Ukraine	4
Italy	4
West Germany	3
Sweden	3
Romania	3
Ireland	3
Germany	3
Zimbabwe	2
New Zealand	2
France	2
Tunisia	1
Suriname	1
Spain	1
Poland	1
Korea, South	1
Costa Rica	1
Bulgaria	1
Brazil	1
Belgium	1

Tableau des épreuve

```
sqlite> SELECT Event, COUNT(*) AS Nombre_Medailles_Or
   ...> FROM Medals
   ...> WHERE Country = 'France'
   ...>   AND Medal = 'Gold'
   ...> GROUP BY Event
   ...> ORDER BY Nombre_Medailles_Or DESC;
```

Event	Nombre_Medailles_Or
épée team	20
football	17
handball	15
foil team	14
team	9
sabre team	6
lightweight coxless four (4-)	4
Tornado - Multihull	4
Team Pursuit (4000m)	4
épée individual	3
foil individual	3
cross-country	3
Olympic Sprint	3
sprint	2
sabre individual	2
pole vault	2
individual	2
double sculls (2x)	2
coxless pair (2-)	2
C-2 (canoe double)	2
C-1 (canoe single)	2
470 - Two Person Dinghy	2
400m	2
+ 100kg (heavyweight)	2
uneven bars	1
points race	1
individual road race	1
individual (FITA Olympic round - 70m)	1
board (lechner)	1
board (Mistral)	1
Keirin	1
K-1 (kayak single)	1
73 - 81kg (half-middleweight)	1
65 - 71kg (lightweight)	1
60 - 66kg	1

TP2 LES PRIX NOBELS

1)

```
sqlite>  SELECT * FROM
   ...> nobel LIMIT 10;
```

Id	Year	Category	Prize	Motivation	PrizeShare	LaureateID	LaureateType	FullName	BirthDate	BirthCity	BirthCountry	Sex	OrganizationName	OrganizationCity	OrganizationCountry	DeathDate	DeathCity	DeathCountry
1	1901	Chemistry	The Nobel Prize in Chemistry 1901	"in recognition of the extraordinary services he has rendered by the discovery of ...	1/1	160	Individual	Jacobus Henricus van 't Hoff	1852-08-30	Rotterdam	Netherlands	Male	Berlin University	Berlin	Germany	1911-03-01	Berlin	Germany
2	1901	Literature	The Nobel Prize in Literature 1901	"in special recognition of his poetic composition, which gives ...	1/1	569	Individual	Sully Prudhomme	1839-03-16	Paris	France	Male	NULL	NULL	NULL	1907-09-07	Châtenay	France
3	1901	Medicine	The Nobel Prize in Physiology or Medicine 1901	"for his work on serum therapy, especially its application against ...	1/1	293	Individual	Emil Adolf von Behring	1854-03-15	Hansdorf (Lawice)	Prussia (Poland)	Male	Marburg University	Marburg	Germany	1917-03-31	Marburg	Germany
4	1901	Peace	The Nobel Peace Prize 1901	NULL	1/2	462	Individual	Jean Henry Dunant	1828-05-08	Geneva	Switzerland	Male	NULL	NULL	NULL	1910-10-30	Heiden	Switzerland
5	1901	Peace	The Nobel Peace Prize 1901	NULL	1/2	463	Individual	Frédéric Passy	1822-05-20	Paris	France	Male	NULL	NULL	NULL	1912-06-12	Paris	France
6	1901	Physics	The Nobel Prize in Physics 1901	"in recognition of the extraordinary services he has rendered by the discovery of ...	1/1	1	Individual	Wilhelm Conrad Röntgen	1845-03-27	Lennep (Remscheid)	Prussia (Germany)	Male	Munich University	Munich	Germany	1923-02-10	Munich	Germany
7	1902	Chemistry	The Nobel Prize in Chemistry 1902	"in recognition of the extraordinary services he has rendered by his work on ...	1/1	161	Individual	Hermann Emil Fischer	1852-10-09	Euskirchen	Prussia (Germany)	Male	Berlin University	Berlin	Germany	1919-07-15	Berlin	Germany
8	1902	Literature	The Nobel Prize in Literature	"the greatest living master of the art of	1/1	571	Individual	Christian Matthias Theodor	1817-11-30	Garding	Schleswig (Germany)	Male	NULL	NULL	NULL	1903-11-01	Charlottenburg	Germany

2)a)

```
sqlite> SELECT DISTINCT Category
   ...> FROM nobel;
```

Category
Chemistry
Literature
Medicine
Peace
Physics
Economics

2)b)

```
sqlite> SELECT DISTINCT FullName, Year
   ...> FROM nobel
   ...> WHERE BirthCountry = 'France'
   ...>    OR OrganizationCountry = 'France'
   ...> ORDER BY FullName ASC;
```

FullName	Year
Alain Aspect	2022
Albert Fert	2007
Alexis Carrel	1912
Alfred Kastler	1966
Alfred Werner	1913
Anatole France	1921
André Frédéric Cournand	1956
André Lwoff	1965
André Paul Guillaume Gide	1947
Anne L'Huillier	2023
Annie Ernaux	2022
Antoine Henri Becquerel	1903
Aristide Briand	1926
Charles Edouard Guillaume	1920
Charles Jules Henri Nicolle	1928
Charles Louis Alphonse Laveran	1907
Charles Robert Richet	1913
Claude Cohen-Tannoudji	1997
Emmanuelle Charpentier	2020
Esther Duflot	2019
Ferdinand Buisson	1927
François Jacob	1965
François Mauriac	1952
Françoise Barré-Sinoussi	2008
Frédéric Joliot	1935
Frédéric Mistral	1904
Frédéric Passy	1901
Gabriel Lippmann	1908
Georges Charpak	1992
Gerard Debreu	1983
Gérard Mourou	2018
Henri Bergson	1927
Henri Moissan	1906
Ilya Ilyich Mechnikov	1908
Irène Joliot-Curie	1935
Jacques Monod	1965
Jean Baptiste Perrin	1926
Jean Dausset	1980
Jean Tirole	2014
Jean-Marie Gustave Le Clézio	2008
Jean-Marie Lehn	1987
Jean-Paul Sartre	1964
Jean-Pierre Sauvage	2016
Jules A. Hoffmann	2011
Louis Eugène Félix Néel	1970
Louis Renault	1907
Luc Montagnier	2008

2)d)

```
sqlite> SELECT COUNT(DISTINCT FullName)
   ...> FROM nobel
   ...> WHERE Sex = 'Female';
```

COUNT(DISTINCT Fu...
67

2)e)

```
sqlite> SELECT FullName, Category, Year
   ...> FROM nobel
   ...> WHERE Sex = 'Female'
   ...>   AND BirthCountry = 'France'
   ...>   AND Year >= 2005;
```

FullName	Category	Year
Françoise Barré-Sinoussi	Medicine	2008
Emmanuelle Charpentier	Chemistry	2020
Annie Ernaux	Literature	2022
Esther Duflot	Economics	2019
Anne L'Huillier	Physics	2023

```
sqlite>
```

2)f)

```
sqlite> SELECT FullName, BirthDate
   ...> FROM nobel
   ...> WHERE LaureateType = 'Individual'
   ...>   AND DeathDate IS NULL
   ...> ORDER BY BirthDate ASC;
```

FullName	BirthDate
Venkatraman Ramakrishnan	NULL
Saul Perlmutter	NULL
Saul Perlmutter	NULL
Rainer Weiss	NULL
Barry C. Barish	NULL
Kip Thorne	NULL
Arthur Ashkin	NULL
Gérard Mourou	NULL
Donna Strickland	NULL
James Peebles	NULL
Michel Mayor	NULL
Didier Queloz	NULL
Roger Penrose	NULL
Reinhard Genzel	NULL
Andrea Ghez	NULL
Syukoro Manabe	NULL
Klaus Hasselmann	NULL
Giorgio Parisi	NULL
Alain Aspect	NULL
John Clauser	NULL
Anton Zeilinger	NULL
Anton Zeilinger	NULL
Jacques Dubochet	NULL
Joachim Frank	NULL
Richard Henderson	NULL
Frances Arnold	NULL
George Smith	NULL
Gregory Winter	NULL
John Goodenough	NULL
Stanley Whittingham	NULL
Akira Yoshino	NULL
Emmanuelle Charpentier	NULL
Jennifer Doudna	NULL

2)g)

```
sqlite> SELECT DISTINCT FullName
   ...> FROM nobel
   ...> WHERE FullName LIKE '%Curie%';
```

FullName
Pierre Curie
Marie Curie, née Sklodowska
Irène Joliot-Curie

3)a)

```
sqlite> INSERT INTO nobel (FullName, Category, Year, Sex, LaureateType)
   ...> VALUES ('Majolie', 'Francais', 2026, 'Male', 'Individual');
```

3) b)

```
sqlite> SELECT FullName
   ...> FROM nobel
   ...> WHERE BirthCity = 'Paris';
```

FullName
Sully Prudhomme
Frédéric Passy
Antoine Henri Becquerel
Pierre Curie
Henri Moissan
Charles Louis Alphonse Laveran
Charles Robert Richet
Léon Victor Auguste Bourgeois
Anatole France
Henri Bergson
Ferdinand Buisson
Frédéric Joliot
Irène Joliot-Curie
André Paul Guillaume Gide
Léon Jouhaux
André Frédéric Cournand
Jean-Paul Sartre
Jacques Monod
Luis F. Leloir
Seán MacBride
Maurice Allais
Pierre-Gilles de Gennes
Françoise Barré-Sinoussi
Patrick Modiano
Jean-Pierre Sauvage

3) c)

```
sqlite> SELECT FullName, Year
   ...> FROM nobel
   ...> WHERE BirthCountry = 'France'
   ...>   AND Category = 'Chemistry';
```

FullName	Year
Henri Moissan	1906
Victor Grignard	1912
Paul Sabatier	1912
Alfred Werner	1913
Frédéric Joliot	1935
Irène Joliot-Curie	1935
Luis F. Leloir	1970
Jean-Marie Lehn	1987
Jean-Marie Lehn	1987
Jean-Pierre Sauvage	2016
Emmanuelle Charpentier	2020
Moungi Bawendi	2023

TP3 LES JOUEURS DE BASKET EVOLUANT EN NBA

1)

```
sqlite> SELECT * FROM joueurs_nba;
```

Name	Team	Number	Position	Age	Height	Weight
Steven Adams	Houston Rockets	12	C	32	211	120
Bam Adebayo	Miami Heat	13	C	28	206	116
Ochai Agbaji	Toronto Raptors	30	SF	25	196	98
Santi Aldama	Memphis Grizzlies	7	C	24	213	98
Trey Alexander	New Orleans Pelicans	23	SG	23	193	84
Nickeil Alexander-Walker	Atlanta Hawks	7	SG	27	196	93
Grayson Allen	Phoenix Suns	8	SG	29	193	90
Jarrett Allen	Cleveland Cavaliers	31	C	27	206	110
Jose Alvarado	New Orleans Pelicans	15	G	27	183	81
Kyle Anderson	Utah Jazz	2	PF	31	206	104
Giannis Antetokounmpo	Milwaukee Bucks	34	F	30	211	110
Thanasis Antetokounmpo	Milwaukee Bucks	43	SF	33	201	99
Cole Anthony	Milwaukee Bucks	50	PG	25	188	84
OG Anunoby	New York Knicks	8	SF	28	201	109
Deni Avdija	Portland Trail Blazers	8	SG	24	206	109
Deandre Ayton	Los Angeles Lakers	5	C	27	213	114
Marvin Bagley III	Washington Wizards	35	PF	26	208	107
Ace Bailey	Utah Jazz	19	SF	19	201	91
Patrick Baldwin, Jr.	Los Angeles Clippers	23	SG	22	206	100
LaMelo Ball	Charlotte Hornets	1	G	24	201	82
Lonzo Ball	Cleveland Cavaliers	2	PG	27	198	86
Paolo Banchero	Orlando Magic	5	C	22	208	114
Desmond Bane	Orlando Magic	3	SG	27	196	98
Dominick Barlow	Philadelphia Sixers	25	F	22	203	98
Harrison Barnes	San Antonio Spurs	40	SF	33	203	102
Scottie Barnes	Toronto Raptors	4	SF	24	201	108
Brooks Barnhizer	Oklahoma City Thunder	23	SF	24	198	104
R.J. Barrett	Toronto Raptors	9	SG	25	198	97
Charles Bassey	Atlanta Hawks	28	PF	24	208	104
Tamar Bates	Denver Nuggets	7	G	23	196	89
Jamison Battle	Toronto Raptors	77	SF	24	201	100
Nicolas Batum	Los Angeles Clippers	33	F	36	203	104
Bradley Beal	Los Angeles Clippers	0	SG	32	193	94
Reece Beekman	Orlando Magic	4	PG	23	191	86
Joan Beringer	Minnesota Timberwolves	19	C	18	208	104
Saddiq Bey	New Orleans Pelicans	41	F	26	201	98
Goga Bitadze	Orlando Magic	35	C	26	208	114
Bismack Biyombo	San Antonio Spurs	18	PF	33	203	116
Anthony Black	Orlando Magic	0	G	21	201	91
Bogdan Bogdanovic	Los Angeles Clippers	10	SG	33	196	102
Adem Bona	Philadelphia Sixers	30	C	22	208	107
Devin Booker	Phoenix Suns	1	SG	28	198	94
Chris Boucher	Boston Celtics	25	PF	32	206	91
Jamaree Bouyea	Milwaukee Bucks	15	G	26	188	82
Tony Bradley	Indiana Pacers	13	C	27	211	113
Malaki Branham	Washington Wizards	8	GF	22	193	82
Christian Braun	Denver Nuggets	0	SF	24	198	100
Koby Brea	Phoenix Suns	14	GF	23	201	98
Mikal Bridges	New York Knicks	25	SG	29	198	95
Miles Bridges	Charlotte Hornets	0	SF	27	201	102
Malcolm Brogdon	New York Knicks	15	SG	32	193	104
Dillon Brooks	Phoenix Suns	3	F	29	198	102
Garrison Brooks	New Orleans Pelicans	NULL	C	26	208	109

2)

```
sqlite> select avg(Age) from joueurs_nba where Team = 'Los Angeles Lakers';
```

avg(Age)
26.142857142857142

Ce code calcule l'âge moyen des joueurs de l'équipe Los Angels Lakers dans la table joueurs_nba.

3) a)

```
sqlite> SELECT MAX(Height) FROM joueurs_nba WHERE Team = 'Chicago Bulls';
```

MAX(Height)
211

3)b)

```
sqlite> SELECT MIN(Weight) FROM joueurs_nba WHERE Position = 'SG';
```

MIN(Weight)
75

3)c)

```
sqlite> SELECT Name, Age, Height, Team
   ...> FROM joueurs_nba
   ...> WHERE Height > 210
   ...> ORDER BY Age ASC
   ...> LIMIT 5;
```

Name	Age	Height	Team
Khaman Maluach	19	218	Phoenix Suns
Ulrich Chomche	20	211	Toronto Raptors
Yang Hansen	20	216	Portland Trail Blazers
Alex Sarr	20	213	Washington Wizards
Rocco Zikarsky	20	221	Minnesota Timberwolves

```
sqlite> SELECT * FROM joueurs_nba
   ...> WHERE Name LIKE '%Wembanyama%';
```

Name	Team	Number	Position	Age	Height	Weight
Victor Wembanyama	San Antonio Spurs	1	F	21	221	107

4)d)

```
sqlite> INSERT INTO joueurs_nba (Name, Team, Number, Position, Age, Height, Weight)
   ...> VALUES ('Majolie', 'Boston Celtics', 99, 'PG', 19, 170, 58);
sqlite> SELECT * FROM joueurs_nba WHERE Name = 'Majolie';
```

Name	Team	Number	Position	Age	Height	Weight
Majolie	Boston Celtics	99	PG	19	170	58

4)e)

```
sqlite> UPDATE joueurs_nba
   ...> SET Team = 'San Antonio Spurs', Number = 75
   ...> WHERE Name LIKE '%Batum%';
sqlite> SELECT * FROM joueurs_nba WHERE Name = 'Nicolas Batum';
```

Name	Team	Number	Position	Age	Height	Weight
Nicolas Batum	San Antonio Spurs	75	F	36	203	104

4)f)

```
sqlite> SELECT Name
   ...> FROM joueurs_nba
   ...> WHERE Team = 'Gotham City';
```

Name
Batman
Robin
Wonder Woman

```
sqlite> DELETE FROM joueurs_nba
   ...> WHERE Team = 'Gotham City';
```